Specific heat of Solids

Aim

To calculate the specific heat capacities of various materials by dipping them in water and measuring the change in temperature.

Theory

The change in temperature of a body and the amount of heat supplied to it are related by the given equation

$$\Delta Q = cm\Delta T$$

where m is the mass of the body and c is a constant called the specific heat of the given substance.

Analysis

The points to ponder asks how one can calculate the water equivalant of the dewar if it were not given. We shall first calculate and verify the value of the water equivalent of the dewar.

Reading	Initial Temperature	Final Temperature	Mass in g
1	62.5	50	59.7
2	55	52	37.1
3	69	61	48.5

In this table, only the first reading is valid because the readings were taken in quick sucession without cooling down the dewar So the temperature T_0 is not known from the second reading onwards. For the first reading T_0 is just the room temperature(the dewar was sitting there for a long time) To calculate the mass equivalant we replace the dewar with m_f grams of water. As the total amount of heat in the system did not change, we can wrie the following equation

$$\Delta Q = c(m_w)(T_i - T_0) = c(m_w + m_f)(T_f - T_0)$$

On simplifing the equation becomes

$$m_f = \frac{m_w(T_i - T_f)}{T_f - T_0}$$

Calculating the value for the first reading we get the value as 28.7 which is within the error range. Because the least count of the thermometer was $1°C$ we see that if the values were 61.5 and 51.0(the worst cases) then the value is 23.3 which is close to the actual value of 23. If this was performed perfectly and not in a hurry the methord looks like this.

Take water at a high temperature(T_i) and pour it into the dewar which is initially kept a long time at room temperature(T_0) and then measure the temperature of the water after pouring it into the dewar(T_f). It would be good to know the mass of water that is being poured into the dewar, but if one can measure it later by measuring the mass of dewar with the water and then subtracting the mass of the dewar. All values are known so now one can calculate m_f Alternatively, one can pour water two times, measure all the values from before both times and then solve for the value of m_f because we have two equations and two variables. This is a bit convoluted but does not involve assuming that the dewar is at room temperature but one can actually measure for the dewar-water combo after water is poured the first time.

Now for calculating values of specific heat for the given materials, we see the table below

	Mass in g		T in steam		T in dewar	
Material	Shots	Water	top	bottom	Initial	Final
Acrylic	41.7	181.6	95.5	-	22.5	26
Brass	144.3	181.1	96.6	97.9	23	28
Iron	134.1	182.7	95.1	97.2	24	29
Ceramic	24.1	180.2	96.6	-	24	25.5

In the table above, top and bottom represent the position of the thermometer in the steam chamber. Top is the temperature of the shots at the top of the heating apparatus and Bottom is the temperature of the shots at the bottom. This happened because the steam arriving hits the bottom and heats up the shots at the bottom more than the shots at the top. The reading of bottom for Acrylic and Ceramic are not present because I could not shove the thermometer deep enough. For brass and iron, the temperature value for T_s can be taken as the average of the values of top and bottom. This is a reasonable approximation as the values are close

The heat lost by the shots is the same as the heat gained by the water-dewar system. We calculate the value of specific heat of the material of the shots by the followig formula

$$c_s = c_w \frac{(m_w + m_f)(T_f - T_i)}{m_s(T_s - T_f)}$$

Doing the caluculation for all the materials we get the values of specific heat as follows

$$\begin{aligned} c_{\text{acrylic}} &= 4.19 \cdot \frac{(181.6+23)(26-22.5)}{41.7(95.5-26)} \\ &= 1.035 \\ c_{\text{brass}} &= 4.19 \cdot \frac{(181.1+23)(28-23)}{144.3(97.2-28)} \\ &= 0.428 \\ c_{\text{iron}} &= 4.19 \cdot \frac{(182.7+23)(29-24)}{134.1(96.1-29)} \\ &= 0.479 \\ c_{\text{ceramic}} &= 4.19 \cdot \frac{(180.2+23)(25.5-24)}{24.1(96.6-25.5)} \\ &= 0.745 \end{aligned}$$

While measuring values, only one measurement was taken, therefore the computable error is due to the least count of the instruments. For the value of T_s, for the values that we have top and bottom measurement the error is the highest value for the following relation

$$\sigma_{T_s} = \frac{\text{bottom} - \text{top}}{2}$$

The largest deviation was for brass, so top and bottom measurements are taken from it. The rest get the values of least count. Hence, the uncertainties are

$$\begin{aligned} \sigma_{m_w} &= 0.1g \\ \sigma_{m_s} &= 0.1g \\ \sigma_{T_i} &= 0.5^\circ C \\ \sigma_{T_f} &= 0.5^\circ C \\ \sigma_{T_s} &= 0.7^\circ C \end{aligned}$$

If f is a function of x_is then the uncertainty in f is given by

$$\sigma_f^2 = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \sigma_{x_i} \right)^2$$

c_s is a function of m_w, T_f, T_i, T_s and m_s.

$$\frac{\partial f}{\partial m_w} = \frac{c_w(T_f - T_i)}{m_s(T_s - T_f)} \sigma_{m_w}$$

$$\frac{\partial f}{\partial T_i} = -\frac{c_w(m_w + m_f)}{m_s(T_s - T_f)} \sigma_{T_i}$$

$$\frac{\partial f}{\partial T_f} = \frac{c_w(m_w + m_f)(T_s - T_i)}{m_s(T_s - T_f)^2} \sigma_{T_f}$$

$$\frac{\partial f}{\partial T_s} = -\frac{c_w(m_w + m_f)(T_f - T_i)}{m_s(T_s - T_f)^2}\sigma_{T_s}$$
$$\frac{\partial f}{\partial m_s} = -\frac{c_w(m_w + m_f)(T_f - T_i)}{m_s^2(T_s - T_f)}\sigma_{m_s}$$

The error in the values is equal to σ_f as there is only one reading Calculating the values of uncertainty using the formula above gives us the error in measurement of each material

$$\sigma_{acrylic} = 0.215$$
$$\sigma_{brass} = 0.063$$
$$\sigma_{iron} = 0.0705$$
$$\sigma_{ceramic} = 0.355$$

Result

The value of the specific heat is reported as $c_x \pm \sigma_x$. The units are same as the units of c_w which is $J/g°C$. Thus, the values of the specific heat of the measured materials are listed below

$$\text{Acrylic} = 1.035 \pm 0.215\ J/g°C$$
$$\text{Brass} = 0.428 \pm 0.063\ J/g°C$$
$$\text{Iron} = 0.479 \pm 0.0705\ J/g°C$$
$$\text{Ceramic} = 0.745 \pm 0.355\ J/g°C$$